Metabolic Syndrome:

Pathophysiology, Epidemiology in Africa, and A Strategic Review of Pharmacological and Complementary Therapeutics

Metabolic Syndrome Crisis

Metabolic Syndrome (MS) is a complex cluster of interrelated metabolic and physiological abnormalities. These disorders, when clustered, substantially increase an individual's lifetime risk for developing chronic conditions, including Type 2 Diabetes Mellitus (T2DM) and premature Cardiovascular Disease (CVD), notably ischemic heart disease.[1] The historical evolution of this concept has moved the clinical focus beyond treating individual risk factors—such as hyperglycemia, dyslipidemia, and hypertension—to recognizing the synergistic and multiplicative effects of their co-occurrence. This clustering underscores the syndrome's multi-systemic nature, necessitating a holistic and integrated therapeutic approach.[1]

Pathophysiological Drivers:

The core pathogenesis of Metabolic Syndrome is centered on the interplay between **Insulin Resistance (IR)** and chronic, low-grade **inflammation**. Insulin is defined by the weakened physiological effect of insulin in the body, where target cells, primarily in the skeletal muscles, liver, and adipose tissues, become less sensitive to the hormone. Consequently, relatively greater amounts of insulin are required to maintain a state of normal blood glucose level.[5] A secondary, yet essential, component driving MS is **visceral obesity**.[1] Adipose tissue, particularly visceral fat, is not merely a passive energy storage depot but functions actively as a dynamic endocrine organ[5] by secreting numerous proinflammatory factors known as adipokines.[5] These inflammatory cytokines interfere with and actively inhibit the insulin signaling pathway within peripheral tissues, including skeletal muscles, the liver, and adipose cells, thereby exacerbating the existing insulin resistance.[2] This establishes a self-amplifying positive feedback

loop: visceral obesity leads to chronic inflammation, which in turn deepens insulin resistance, resulting in further metabolic dysfunction, such as elevated triglycerides (TG). **This means that therapeutic strategies aiming to halt MS progression must prioritize addressing insulin resistance and chronic inflammation rather than focusing solely on managing downstream symptoms like elevated blood pressure or glucose levels**. Intervention focusing on weight management is therefore a crucial mechanistic break point. The accumulated adverse effects of this chronic inflammatory state and resulting metabolic dysfunction manifest across several body systems.[2] Specifically, endothelial dysfunction and hypertension, both consequential outcomes of MS, can ultimately lead to ischemic heart disease.[1] Other non-core underlying risk factors that contribute to the development of MS include **physical inactivity, aging, hormonal imbalance, and inherent genetic or ethnic predispositions**.[1]

Diagnostic Criteria

The clinical diagnosis of Metabolic Syndrome relies on defined quantitative criteria, typically requiring the presence of any three out of five specific metabolic risk factors, to accurately predict future T2DM and CVD risk. The lack of a single, universally accepted definition results in diagnostic friction, as different criteria proposed by major organizations—such as the National Cholesterol Education Program Adult Treatment Panel III (NCEP-ATP III), the International Diabetes Federation (IDF), and the Joint Interim Statement (JIS)—lead to varying prevalence estimates in populations.[1] The **NCEP-ATP III Criteria** requires meeting 3 of the 5 criteria. **These criteria include elevated waist circumference (as a measure of central obesity), elevated triglycerides (TG), reduced High-Density Lipoprotein Cholesterol (HDL-C), elevated blood pressure (BP), and elevated fasting glucose** [1, 5].

Parameter	NCEP-ATP III (2004)	IDF (2005) (Europid)	African Optimal WC Recommendations (Ranges)
Central Obesity (WC)	Men: ≥ 102 cm / Women: ≥ 88 cm	Required: Men: ≥ 94 cm / Women: ≥ 80 cm	Men: ≥ 86 cm / Women: 80.3–92 cm
Triglycerides (TG)	≥ 150 mg/dL (1.7 mmol/L) or drug treatment	≥ 150 mg/dL (1.7 mmol/L) or drug treatment	≥ 150 mg/dL (1.7 mmol/L) or drug treatment
HDL Cholesterol (HDL-C)	Men: < 40 mg/dL / Women: < 50 mg/dL or drug treatment	Men: < 40 mg/dL / Women: < 50 mg/dL or drug treatment	Men: < 40 mg/dL / Women: < 50 mg/dL or drug treatment
Blood Pressure (BP)	$\geq 130/85$ mmHg or drug treatment	$\geq 130/85$ mmHg or drug treatment	$\geq 130/85$ mmHg or drug treatment
Fasting Glucose (FPG)	≥ 100 mg/dL or drug treatment	≥ 100 mg/dL or drug treatment	≥ 100 mg/dL or drug treatment

The Challenge of Anthropometric Definitions in African Populations

A significant challenge in standardizing MS diagnosis globally is the application of anthropometric definitions across different ethnic groups. The established NCEP-ATP III and generalized IDF WC cut-offs, often derived from Caucasian populations, may inaccurately reflect the true visceral adiposity risk in African populations due to inherent differences in body composition, lean mass distribution, and fat patterning.[8] Research attempting to establish African-specific criteria has highlighted considerable complexity and heterogeneity. One study, using the JIS criteria as a basis, found optimal Waist Circumference cut-off points for predicting the presence of at least two other MS components to be **86 cm for men** (with a sensitivity of 61.2% and specificity of 82.9%) and **92 cm for women** (with a sensitivity of 45.9% and specificity of 81.9%).[7] However, this finding contrasts sharply with another analysis in a different cohort that suggested an optimal threshold of **80.3 cm for women** aged 18–49 years, a value that aligns more closely with standard IDF guidance for Europids.[8]

This contradiction in optimal anthropometric thresholds—for example, 92 cm versus 80.3 cm for women—creates significant diagnostic friction and methodological bias when comparing regional prevalence studies. Without a consistent, validated measure of central obesity, prevalence tracking,

resource allocation, and therapeutic guidelines based on visceral fat risk are scientifically compromised. Consequently, for reliable public health planning across the continent, there must be a unified, high-priority effort to establish large-scale, pan-African anthropometric criteria that accurately reflect regional body composition profiles.

Metabolic Syndrome in Africa Population

The continent of Africa faces a substantial and rapidly growing burden of Metabolic Syndrome. An analysis compiling 345 prevalence data points from 29 countries revealed that the overall pooled prevalence of MS in Africa is **32.4%** (95% CI: 30.2–34.7).[9] This statistic indicates that nearly one in every three adults across the continent is affected by this condition. Geographically, the burden is not uniformly distributed. The Southern African region was identified as the most affected sub-region, reporting a prevalence of **33.6%** (95% CI: 28.3–39.1). This was closely followed by populations residing in upper-middle-income economies, suggesting that economic development and the resulting changes in diet and lifestyle are key risk modifiers.[9] A defining characteristic of MS epidemiology in Africa is the strong female predominance. The pooled prevalence of MS in women across Africa is significantly higher at **36.9%** (95% CI: 33.2–40.7), compared to men at **26.7%** (95% CI: 23.1–30.5) ($P < 0.001$).[9] This disparity is even more pronounced in specific regional studies; for example, in North-Western Nigeria, the prevalence was reported as 42.83% for females versus 27.36% for males.[10] Analysis of the component profile reveals that African women frequently present with low HDL cholesterol (up to 62.2%) and central obesity (up to 49.8%) as the most prevalent components of the syndrome.[10] The concentration of central obesity and low HDL in women suggests a unique combination of biological (hormonal factors, parity) and socioeconomic influences, such as changes in gendered societal roles, dietary habits associated with urbanization, and potential genetic predispositions for central fat deposition. This demographic pattern dictates that public health campaigns must be highly targeted and culturally sensitive, prioritizing weight and lipid profile interventions for women. Furthermore, the prevalence of MS shows a clear dependence on age. The overall pooled estimate for adults (Age > 18 years) was 33.1%, significantly higher than the 13.3% observed in children (0–18 years).[9] This increase can be rapid, with some cohorts demonstrating prevalence rising from 11% in subjects aged 20–29 years to as high as 89% in those aged 70–79 years.[4] Although prevalence continues to increase with age, the available data suggest that adults classified as middle-aged (40–60 years) are predominantly affected.[4]

Across the general population, the most common components observed in a study of North-Western Nigeria were low HDL (56.1%) and hypertension (46.1%).[10] Conversely, while central obesity and low HDL were commoner in females, combinations involving high blood pressure and hypercholesterolemia often demonstrated male predominance.[4]

Pharmacological Management: Targets, Toxicity, and Therapeutic Efficacy of Orthodox Drugs

Core Therapeutic Principle: Lifestyle First, Pharmacotherapy Second: The primary intervention for Metabolic Syndrome remains comprehensive lifestyle modification, including **intentional weight loss, increased physical activity,** and **dietary change**.[1] Pharmacological management is introduced when the **underlying metabolic risk factors fail to respond adequately to these non-drug interventions**. The therapeutic strategy for MS is component-specific, focusing on independently managing **atherogenic dyslipidemia, arterial hypertension, and hyperglycemia/insulin resistance**.[13]

Orthodox Management of Atherogenic Dyslipidemia

Atherogenic dyslipidemia, characterized by elevated triglycerides (TG), elevated apolipoprotein B, small LDL particles, and low HDL cholesterol (HDL-C), is a core risk factor.[1]

Statins: These 3-hydroxy-3-methyl-glutaryl-coenzyme A (HMG-CoA) reductase inhibitors represent the cornerstone of dyslipidemia management.[13] They are crucial for lowering LDL cholesterol (LDL-C) and are available in various intensity regimens (low, moderate, and high) depending on the specific drug and dose (e.g., atorvastatin, simvastatin, rosuvastatin).[13] While Statins are generally well-tolerated, but the primary clinical concern remains **musculoskeletal risk**, ranging from mild myalgia to, in rare cases, severe rhabdomyolysis.[15]

Non-Statin Therapies: These are often used when LDL-C goals are not met or in cases of statin intolerance. Ezetimibe works by inhibiting cholesterol absorption in the small intestine via the Niemann-Pick C1-Like-1 (NPC1L1) transporter.[13] Fibrates (e.g., fenofibrate, gemfibrozil) and nicotinic acid are

useful primarily for managing severe hypertriglyceridemia (TG ≥ 500 mg/dL).[13] Newer classes, such as Proprotein convertase subtilisin/kexin type 9 (PCSK9) inhibitors (e.g., alirocumab, evolocumab), dramatically lower LDL-C by preventing the degradation of LDL receptors.[13]

Alternative Therapy Management of Atherogenic Dyslipidemia (Panaceutics):

Petriverol® is our flagship Tri-herbal anti-hypercholestrolemic formulation synergistically functioning to block Niemann-Pick C1-Like-1 (NPC1L1) transporter, repress hepatic HMG-CoA reductase gene expression even in low-fat diet and dual protein inhibition and gene repression of intestinal G-protein coupled receptor 146 (GPR146).

Orthodox Management of Hyperglycemia and Insulin Resistance

The management of dysglycemia, particularly insulin resistance, is central to halting MS progression. **Metformin** is a primary therapeutic agent for addressing IR. It operates primarily by activating Adenosine Monophosphate-Activated Protein Kinase (AMPK), which subsequently reduces hepatic glucose production.[14] While Metformin is highly valued for its efficacy in improving insulin sensitivity but is frequently associated with **gastrointestinal intolerance** (e.g., diarrhea, nausea), **lactic acidosis (LA)**, and **Vitamin B12 deficiency**, linked to the malabsorption of Vitamin B12.[17] This is connected to a decline in cognitive function and an increased chance of developing depressive symptoms.[17] Nowadays, **Incretin-Based Therapies** that enhance insulin sensitization or augment postprandial insulin secretion are also available.[12] Examples of the drugs include **GLP-1 Receptor Agonists (GLP-1 RAs) and DPP-4 Inhibitors,** which enhance insulin secretion, slowing gastric emptying, and promoting weight reduction.[l] It is worthy of note that **DPP-4 Inhibitors** typically achieve a two-fold increase in the postprandial plasma levels of endogenous GLP-1.[12] **Thiazolidinediones (TZDs)** such as rosiglitazone, function as PPAR-gamma agonists, effectively increasing insulin sensitization in peripheral tissues.[12]

Alternative Therapy Management of Hyperglycemia and Insulin Resistance

(Panaceutics):

Sphenolax® is our flagship Penta-Herbal formulation with TGR5 agonism, DPP-IV inhibition, Alpha-amylase inhibition, dual DYRK1A/GSK-3β inhibition and selective AMP-activated protein kinase (AMPK) activation. Sphenolax® slowly repairs adult pancreas in both human and mouse model of diabetes, providing highly sustained pancreatic beta-cell function.

Orthodox Management of Arterial Hypertension

Hypertension is a highly prevalent component of MS in African cohorts, necessitating robust management strategies.[10] Five major drug classes are available as first-line antihypertensive treatments: **Thiazide/thiazide-like diuretics, Calcium Channel Blockers (CCBs), Angiotensin-Converting Enzyme Inhibitors (ACEi), Angiotensin Receptor Blockers (ARBs), and Beta-blockers (BBs).**[13] Given the strong link between MS and cardio-renal risk, blockers of the Renin–Angiotensin System (RAS), specifically ACEi or ARBs, are often preferred and form a common component of general combination treatment strategies.[13]

Alternative Therapy Management of Arterial Hypertension (Panaceutics):

BPVomin® is our flagship Bi-Herbal formulation with renin-angiotensin-aldosterone system (RAAS) modulation via Angiotensin Converting Enzyme I/II inhibition, dual endothelin receptor (endothelin A and endothelin B receptors) antagonism.

Orthodox Management of Obesity

Drugs for obesity management include prescription medications like the gut-acting, gastric and

pancreatic lipases **inhibitor-orlistat, central acting phentermine-topiramate (Qsymia), naltrexone-(mu-Opioid receptor)-bupropion-(**reuptake inhibitor of norepinephrine and dopamine**) (Contrave),** GLP-1 receptor **agonism is the mechanism of action of liraglutide (Saxenda), semaglutide (Wegovy), and tirzepatide (Zepbound). These drugs work by targeting hormones to suppress appetite, reduce fat absorption, or increase feelings of fullness.** Limitations include common side effects like nausea, vomiting, and dizziness, and the fact that some drugs may be limited to individuals with specific genetic disorders or pre-existing conditions like cardiovascular disease**.** GLP-1 Receptor Agonists are effective weight loss and glucose-lowering agents, but their main side effects are highly prevalent gastrointestinal symptoms, including **nausea, vomiting, abdominal pain, and diarrhea**.[18] These effects do not usually lead to treatment discontinuation, and evidence suggests that long-acting formulations may be associated with less nausea than short-acting compounds.[18] A critical, indirect toxicological risk is **acute kidney injury (AKI)**. Severe GI symptoms can lead to dehydration, which, in turn, can cause pre-renal AKI.[18] Therapeutic management in cases of severe overdose or persistent symptoms is primarily supportive, ensuring adequate fluid balance. The association between GLP-1 RAs and pancreatitis has been extensively debated

Alternative Therapy Management of Obesity (Panaceutics):

Nu-beta® is our flagship Poly-Herbal formulation designed for weight management. This product specifically upregulates the expression of Jak3 and Uncoupling Protein 1 (UCP-1) within the adipose tissue in diet-induced obesity. The combination of the mechanisms above positions Nu-beta® as key to transitioning White Adipose Tissue to brown and beige types.

Orthodox Management of acute ischemic stroke

The main drug for treating an acute ischemic stroke is **tissue plasminogen activator (tPA),** a “clot-

dissolving" medication that must be given within 3 to 4.5 hours of symptom onset to break up blood clots. After the acute phase, other drugs, including blood thinners (antiplatelets like aspirin, and anticoagulants) are prescribed for prevention, and medications to manage blood pressure, seizures, and muscle spasms. A major limitation for the use of tPA is not suitable for all patients due to the increased risk of severe bleeding especially in hemorrhagic stroke patients where tPA worsens bleeding.

Alternative Therapy Management of acute ischemic stroke (Panaceutics):

Exigencin® is our flagship Herbal formulation designed as tri-mechanistic agent. First, it is designed as a weak dual Phyto-thrombolytic/phyto-anticoagulant agent. Secondly, it blocks TLR4-Mediated Inflammation thus preventing extensive tissue damage and lastly, Exigencin® initiates Brain-derived neurotrophic factor-mediated neurogenesis within the CNS; forcing repair and the remodeling of the extracellular matrix in preparation for functional recovery.

Pitch for Alternative and Complementary Treatment (ACT) Options

African Flora offers a vast and largely untapped pharmacopeia of botanicals for chronic metabolic disorders treatment. Given the necessity of polypharmacy to manage MS components and the corresponding high risk of pharmacological toxicity and costs associated with orthodox medicine, there is a strong justification for rigorous exploration and integration of alternative and complementary treatment (ACT) options. ACT offers safe alternatives that can reduce morbidity and significantly decrease the overall financial cost associated with treating MS.[20] Indigenous knowledge and traditional treatments hold intrinsic value on the use of botanicals[21] offering treatments often perceived as effectiveness and culturally acceptable within local communities. Nowadays, with the advent of cutting-edge biomedical and pharmaceutical research, **PANACEUTICS** provides yet another layer of evidence such as mechanism of action, phytochemical basis of actions, and No-observed Adverse Effect Level

(NOAELs)-Human-Equivalent Dose (HED)-derived **Dosing Regimen** which is crucial path for patient confidence, acceptance and adherence which were the major limiting factor in managing chronic NCDs in Africa with Traditional African Medicine (TAM). [21] Ultimately, **PANACEUTICS** taps into the incredible millennial-old history and evidence of the utilization of each plant in its herbal portfolio, and systematically integrate into modern health systems through a standardized, collaborative research effort involving ethnobotanists, pharmacologists, and clinicians.

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